

Module 1: The Scientific Method & Basic Chemistry

Comprehensive Identification & Multiple-Choice Reviewer — 82 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (82 ITEMS)

- ____ 1. A systematic way of learning about the natural world by applying rules of logic to how information is gathered and conclusions are drawn is the ____.
- a.) inductive reasoning b.) theory of relativity c.) scientific method d.) anecdotal evidence
- ____ 2. In the scientific method, careful observation of the natural environment typically prompts a ____ that is grounded in existing knowledge.
- a.) proof b.) question c.) conclusion d.) law
- ____ 3. A proposed testable explanation formulated as a possible answer to an observed scientific question is a ____.
- a.) scientific law b.) variable c.) hypothesis d.) control
- ____ 4. To be methodologically valid, a scientific hypothesis must be phrased as a ____, rather than as a question.
- a.) statement b.) riddle c.) inquiry d.) paradox
- ____ 5. A fundamental characteristic of any valid scientific hypothesis is that it must be testable, meaning that it has the potential to be proven ____.
- a.) inviolable b.) infallible c.) false d.) absolute
- ____ 6. In testing a hypothesis, scientists formulate a ____ that specifies the expected outcome and dictates the necessary experimental design.
- a.) paradigm b.) postulate c.) prediction d.) retraction
- ____ 7. An experimental test where all conditions are held constant except for the specific factor under investigation is known as a ____.
- a.) observational drift b.) blind survey c.) anecdotal trial d.) controlled experiment
- ____ 8. In a controlled experiment, the single factor that differs between the experimental group and the control group is the ____.
- a.) variable b.) constant c.) placebo d.) hypothesis
- ____ 9. In a controlled experiment, the cohort that is maintained under standard conditions without receiving the experimental treatment is the ____.
- a.) cohort sample b.) control group c.) experimental group d.) variable group
- ____ 10. In a controlled experiment, the cohort subjected to the specific variable being tested is the ____.
- a.) null group b.) control group c.) experimental group d.) baseline cohort
- ____ 11. The final phase of the scientific method where experimental data is interpreted to accept or reject the hypothesis is drawing a ____.
- a.) hypothesis b.) variable c.) conclusion d.) prediction
- ____ 12. To ensure that experimental findings and differences between groups are not merely the result of chance, researchers evaluate ____.
- a.) random variance b.) subjective consensus c.) anecdotal frequency d.) statistical significance

- ____ 13. Anything that possesses mass, occupies three-dimensional space, and is made up of atoms is defined as ____.
- a.) vacuum b.) matter c.) energy d.) radiation
- ____ 14. The fundamental structural units of matter that cannot be broken down into simpler substances by ordinary chemical means are ____.
- a.) cells b.) macromolecules c.) atoms d.) compounds
- ____ 15. The three subatomic particles that compose atoms are protons, electrons, and ____.
- a.) positrons b.) neutrons c.) quarks d.) photons
- ____ 16. A pure form of matter that consists exclusively of one single type of atom is called an ____.
- a.) isotope b.) element c.) compound d.) mixture
- ____ 17. Of the more than 100 known chemical elements, approximately ____ elements are found in and essential to the human body.
- a.) 50 b.) 92 c.) 4 d.) 20
- ____ 18. The four most abundant elements that make up the vast majority of human body tissues are carbon, oxygen, hydrogen, and ____.
- a.) calcium b.) phosphorus c.) nitrogen d.) iron
- ____ 19. The standard unit of measurement used to quantify the atomic weight or mass of an element is the ____.
- a.) gram per mole b.) joule c.) atomic mass unit d.) angstrom
- ____ 20. The atomic weight of an element is primarily determined by the total number of protons and ____.
- a.) neutrons b.) valence shells c.) photons d.) electrons
- ____ 21. Atoms of the same element that contain the same number of protons but differ in their number of neutrons are called ____.
- a.) isomers b.) isotopes c.) allotropes d.) monomers
- ____ 22. Unstable isotopes that spontaneously break down over time and emit radiation or subatomic particles are called ____.
- a.) ions b.) electrolytes c.) stable isotopes d.) radioisotopes
- ____ 23. A distinct chemical substance formed when two or more different elements chemically combine with properties entirely different from the parent elements is a ____.
- a.) compound b.) colloid c.) mixture d.) suspension
- ____ 24. A strong chemical bond formed when two or more atoms share electrons in their outer valence shells is a ____ bond.
- a.) metallic b.) hydrogen c.) covalent d.) ionic
- ____ 25. A chemical bond resulting from the complete transfer of valence electrons and the subsequent mutual attraction between oppositely charged ions is an ____ bond.
- a.) ionic b.) hydrogen c.) peptide d.) covalent
- ____ 26. Water can dissolve a wide range of biological solutes and exhibits high heat capacity because of the ____ of its molecules.
- a.) viscosity b.) polarity c.) neutrality d.) nonpolar nature
- ____ 27. The weak electrostatic attraction between the slightly positive hydrogen atom of one water molecule and the slightly negative oxygen atom of another is a ____ bond.
- a.) ionic b.) phosphodiester c.) covalent d.) hydrogen
- ____ 28. Water's ability to absorb substantial amounts of heat energy with minimal change in its own temperature is due to its high ____.
- a.) surface tension b.) viscosity c.) vapor pressure d.) heat capacity
- ____ 29. The property that makes sweating an effective physiological cooling mechanism for humans is water's high heat of ____.
- a.) sublimation b.) vaporization c.) fusion d.) crystallization

- _____ 30. Large biological molecules composed of thousands of covalently linked atoms organized into chains are called _____.
a.) macromolecules b.) micelles c.) microelements d.) coenzymes
- _____ 31. A complex macromolecule constructed from repeating, identical or similar molecular subunits joined together is a _____.
a.) dimer b.) monomer c.) polymer d.) monosaccharide
- _____ 32. The individual, small molecular building blocks that link together to assemble polymers are _____.
a.) polymers b.) complexes c.) cofactors d.) monomers
- _____ 33. The chemical reaction that joins monomers together to build polymers by releasing a molecule of water is a _____ reaction.
a.) dehydration b.) oxidation c.) hydrolysis d.) carboxylation
- _____ 34. The chemical reaction that breaks down polymers into individual monomers by consuming a molecule of water is a _____ reaction.
a.) hydrolysis b.) dehydration c.) reduction d.) condensation
- _____ 35. The four major classes of biological macromolecules essential for life are carbohydrates, proteins, nucleic acids, and _____.
a.) vitamins b.) lipids c.) minerals d.) electrolytes
- _____ 36. Biomolecules composed entirely of carbon, hydrogen, and oxygen that serve as the primary source of cellular fuel and energy are _____.
a.) nucleic acids b.) steroids c.) lipids d.) carbohydrates
- _____ 37. The simplest carbohydrate units, consisting of single sugar molecules containing three to seven carbon atoms, are _____.
a.) triglycerides b.) monosaccharides c.) oligosaccharides d.) polysaccharides
- _____ 38. Six-carbon simple sugars like glucose, fructose, and galactose are classified based on their carbon count as _____.
a.) hexoses b.) trioses c.) pentoses d.) heptoses
- _____ 39. The six-carbon monosaccharide abundant in carbohydrates that circulates in the blood as the primary fuel for human cells is _____.
a.) glucose b.) ribose c.) sucrose d.) cellulose
- _____ 40. The hexose monosaccharide naturally found in abundance in fruits and vegetables is _____.
a.) maltose b.) galactose c.) glycogen d.) fructose
- _____ 41. The hexose monosaccharide commonly found as a component of the disaccharide lactose in dairy products is _____.
a.) galactose b.) glycogen c.) ribose d.) starch
- _____ 42. The five-carbon monosaccharide (pentose) that forms an integral structural component of RNA nucleotides is _____.
a.) ribose b.) sucrose c.) glucose d.) fructose
- _____ 43. Short carbohydrate chains consisting of a few monosaccharides joined by dehydration synthesis are called _____.
a.) polysaccharides b.) monomers c.) phospholipids d.) oligosaccharides
- _____ 44. A double sugar formed when two monosaccharides are linked together by a covalent bond is a _____.
a.) disaccharide b.) monosaccharide c.) polysaccharide d.) hexose
- _____ 45. Common household table sugar is a disaccharide chemically known as _____.
a.) lactose b.) cellulose c.) sucrose d.) maltose
- _____ 46. Extensive carbohydrate macromolecules formed by linking numerous monosaccharides into long, complex chains are _____.
a.) disaccharides b.) triglycerides c.) polysaccharides d.) peptides

- ____ 47. The primary storage polysaccharide in plants, consisting of long chains of linked glucose units, is ____.
a.) starch b.) cellulose c.) glycogen d.) chitin
- ____ 48. The primary storage polysaccharide in animals and humans, stored predominantly in the liver and skeletal muscles, is ____.
a.) glycogen b.) sucrose c.) cellulose d.) starch
- ____ 49. The rigid structural polysaccharide found in plant cell walls, commonly present in bamboo, jute, and cotton, is ____.
a.) glucose b.) glycogen c.) starch d.) cellulose
- ____ 50. Water-insoluble organic biomolecules that are hydrophobic due to their nonpolar nature are ____.
a.) proteins b.) carbohydrates c.) nucleic acids d.) lipids
- ____ 51. The three classes of lipids essential to human physiological health are phospholipids, steroids, and ____.
a.) peptides b.) polysaccharides c.) coenzymes d.) triglycerides
- ____ 52. A triglyceride molecule is constructed from one molecule of glycerol chemically bonded to three ____.
a.) amino acids b.) nucleotides c.) fatty acids d.) monosaccharides
- ____ 53. A fatty acid chain that contains no double bonds between carbon atoms and is fully saturated with hydrogens is a ____ fatty acid.
a.) polyunsaturated b.) saturated c.) unsaturated d.) trans
- ____ 54. A fatty acid chain that contains one or more double bonds between carbon atoms is classified as an ____ fatty acid.
a.) saturated b.) solid c.) hydrogenated d.) unsaturated
- ____ 55. Triglycerides serve as critical long-term energy reservoirs in animals because per gram, they yield ____ the energy of carbohydrates.
a.) ten times b.) equal c.) twice d.) half
- ____ 56. Lipid molecules that contain both a hydrophilic polar head and two hydrophobic nonpolar fatty acid tails are described as ____.
a.) hydrophobic b.) nonpolar c.) hydrophilic d.) amphipathic
- ____ 57. The major structural lipid component of biological plasma membranes is the ____.
a.) phospholipid b.) triglyceride c.) glycogen d.) starch
- ____ 58. A class of lipids characterized by a molecular structure composed of four interconnected carbon rings is ____.
a.) phospholipids b.) steroids c.) hexoses d.) triglycerides
- ____ 59. The essential steroid that integrates into plasma membranes and serves as the precursor for sex hormones like estrogen and testosterone is ____.
a.) insulin b.) cholesterol c.) glycerol d.) cellulose
- ____ 60. Polymers composed of specific sequences of 20 different building blocks linked by peptide bonds are ____.
a.) proteins b.) nucleic acids c.) polysaccharides d.) lipids
- ____ 61. The basic building block of all proteins is the ____.
a.) fatty acid b.) nucleotide c.) amino acid d.) monosaccharide
- ____ 62. Every amino acid features a central carbon atom bonded to a hydrogen atom, an amino group, a side chain, and an acidic ____ group.
a.) carboxyl b.) phosphate c.) hydroxyl d.) methyl
- ____ 63. The distinctive portion of an amino acid molecule that gives it unique chemical characteristics and differentiates it from all other amino acids is the ____.
a.) amino group b.) central carbon c.) side chain d.) carboxyl group
- ____ 64. Amino acids that the human body can synthesize endogenously from glucose metabolism are classified as ____ amino acids.
a.) essential b.) exogenous c.) dietary d.) non-essential

- _____ 65. Amino acids that cannot be produced by the human body and must be acquired through dietary sources like meat, eggs, and fish are _____ amino acids.
a.) synthetic b.) essential c.) non-essential d.) endogenous
- _____ 66. The specific, linear sequence of amino acids linked together in a polypeptide chain represents a protein's _____ structure.
a.) primary b.) quaternary c.) tertiary d.) secondary
- _____ 67. The folding of an amino acid chain into alpha-helices or beta-pleated sheets due to hydrogen bonding represents the _____ structure of a protein.
a.) primary b.) secondary c.) quaternary d.) tertiary
- _____ 68. The overall three-dimensional globular conformation of a single polypeptide chain, characteristic of functional enzymes, is its _____ structure.
a.) secondary b.) quaternary c.) tertiary d.) primary
- _____ 69. The complex protein structure produced when two or more individual polypeptide subunits join together, as seen in hemoglobin, is the _____ structure.
a.) secondary b.) primary c.) tertiary d.) quaternary
- _____ 70. Specialized proteins that act as biological catalysts by accelerating chemical reactions without being consumed are _____.
a.) enzymes b.) hormones c.) antibodies d.) receptors
- _____ 71. The specific reactant molecule upon which an enzyme acts during a catalyzed biochemical reaction is the _____.
a.) product b.) substrate c.) inhibitor d.) coenzyme
- _____ 72. The specialized region or pocket on an enzyme where the substrate binds to initiate catalysis is the _____ site.
a.) allosteric b.) regulatory c.) cofactor d.) active
- _____ 73. Nonprotein chemical substances (such as metal ions) required by certain enzymes to facilitate catalytic activity are _____.
a.) substrates b.) cofactors c.) polymers d.) monomers
- _____ 74. Organic cofactors derived from dietary nutrients, such as vitamins, that assist enzymatic reactions are specifically called _____.
a.) coenzymes b.) amphipaths c.) substrates d.) isotopes
- _____ 75. The two varieties of nucleic acids that store and transmit hereditary genetic information in cells are RNA and _____.
a.) DNA b.) NADH c.) GTP d.) ATP
- _____ 76. The fundamental monomeric subunit of nucleic acids, consisting of a five-carbon sugar, a phosphate group, and a nitrogenous base, is a _____.
a.) nucleoside b.) nucleotide c.) fatty acid d.) amino acid
- _____ 77. A distinct segment of DNA along a chromosome that encodes the instructions for the sequence of amino acids in a protein is a _____.
a.) ribosome b.) gene c.) nucleotide d.) codon
- _____ 78. The five-carbon pentose sugar found in DNA nucleotides is _____.
a.) glucose b.) deoxyribose c.) fructose d.) ribose
- _____ 79. The nitrogenous base found uniquely in DNA that pairs with adenine, but is absent in RNA, is _____.
a.) guanine b.) thymine c.) uracil d.) cytosine
- _____ 80. The single-stranded nucleic acid containing ribose sugar and the bases adenine, guanine, cytosine, and uracil is _____.
a.) RNA b.) DNA c.) ATP d.) cAMP
- _____ 81. The nitrogenous base found in RNA that substitutes for thymine and base-pairs with adenine is _____.

a.) uracil b.) guanine c.) thymine d.) cytosine

_____ **82.** In human cells, RNA plays a pivotal role in _____ synthesis as messenger RNA, transfer RNA, and ribosomal RNA.

a.) steroid b.) carbohydrate c.) lipid d.) protein

COMPLETE ANSWER KEY & EXPLANATIONS

Module 1: The Scientific Method & Basic Chemistry — Reference Key with Concept Explanations

Item	Key	Correct Answer	Curriculum Explanation & Reference
1	C	scientific method	[Scientific Method Overview] The scientific method is defined as a logical process for learning about the natural world through systematic observation, hypothesis testing, and data analysis.
2	B	question	[Phase 1: Observation] Phase 1 of the scientific method involves making careful observations which prompt a specific question based on existing scientific knowledge.
3	C	hypothesis	[Phase 2: Hypothesis] A hypothesis is a tentative, testable statement proposed to explain an observed phenomenon.
4	A	statement	[Phase 2: Hypothesis Formulation] A scientific hypothesis must always be formulated as a declarative statement (e.g., 'Daily exposure to the sun leads to increased levels of happiness') rather than a question.
5	C	false	[Phase 2: Falsifiability] A hypothesis can never be proven absolutely true beyond all doubt, but it must be testable so that empirical evidence can prove it false (falsifiability).
6	C	prediction	[Phase 3: Prediction] Making a prediction based on the hypothesis determines the precise experiments and observations required to test the proposed explanation.
7	D	controlled experiment	[Phase 3: Controlled Experiment] A controlled experiment tests a hypothesis by comparing a control group with an experimental group under identical conditions.
8	A	variable	[Phase 3: Experimental Variable] The variable is the specific factor being manipulated or tested to reveal its biological effect.
9	B	control group	[Phase 3: Control Group] The control group serves as the baseline for comparison and does not receive the experimental manipulation.
10	C	experimental group	[Phase 3: Experimental Group] The experimental group receives the variable factor whose effect is being investigated.
11	C	conclusion	[Phase 4: Conclusion] Conclusions are drawn after analyzing and interpreting empirical data gathered from the experiment.
12	D	statistical significance	[Phase 4: Statistical Significance] Statistical significance provides mathematical assurance that observed experimental differences did not occur simply by chance.
13	B	matter	[Matter & Atoms] Matter is defined as anything that has mass and occupies physical space.

14	C	atoms	[Atoms] Atoms are the smallest functional units of an element that retain its chemical properties and cannot be chemically cleaved.
15	B	neutrons	[Subatomic Particles] Atoms consist of positively charged protons, uncharged neutrons (in the nucleus), and negatively charged electrons.
16	B	element	[Elements] An element is a substance containing only one kind of atom and cannot be broken down chemically.
17	D	20	[Elements in the Human Body] While there are over 100 known elements, only about 20 are naturally present in the human body.
18	C	nitrogen	[Major Elements of Human Body] Carbon, Oxygen, Hydrogen, and Nitrogen (COHN) comprise roughly 96% of human body mass.
19	C	atomic mass unit	[Atomic Mass] The atomic mass unit (amu or Dalton) is standardly used to express atomic weight.
20	A	neutrons	[Atomic Weight] Since electrons have negligible mass, atomic mass/weight is almost entirely determined by the sum of protons and neutrons in the nucleus.
21	B	isotopes	[Isotopes] Isotopes are variants of a particular chemical element which differ in neutron number, and consequently in nucleon number.
22	D	radioisotopes	[Radioisotopes] Radioisotopes are radioactive isotopes that decay by emitting ionizing radiation (such as alpha, beta, or gamma rays).
23	A	compound	[Chemical Compounds] A compound is a chemical entity formed by bonding two or more elements, possessing unique chemical characteristics distinct from its components.
24	C	covalent	[Covalent Bonds] Covalent bonds involve the mutual sharing of valence electron pairs between atoms.
25	A	ionic	[Ionic Bonds] Ionic bonds form when one atom donates electrons to another, generating cations and anions that attract each other.
26	B	polarity	[Water Polarity] Polarity refers to the asymmetrical distribution of electrical charge, resulting in partial positive and negative poles in water.
27	D	hydrogen	[Hydrogen Bonds] Hydrogen bonds occur when partially positive hydrogen atoms attract electronegative atoms (like oxygen or nitrogen) in adjacent molecules.
28	D	heat capacity	[Water Heat Capacity] High heat capacity allows water to stabilize body temperature by buffering sudden environmental thermal shifts.
29	B	vaporization	[Heat of Vaporization] High heat of vaporization means evaporating water absorbs significant thermal energy, carrying heat away from the body surface.

30	A	macromolecules	[Macromolecules] Macromolecules are giant biomolecules formed by linking smaller organic subunits into long chains.
31	C	polymer	[Polymers & Monomers] A polymer is a large macromolecule composed of repeated building blocks called monomers.
32	D	monomers	[Monomers] Monomers are the fundamental recurring units that polymerize to form polymers.
33	A	dehydration	[Dehydration Synthesis] Dehydration synthesis (condensation reaction) removes water to covalently bond monomers into a growing polymer.
34	A	hydrolysis	[Hydrolysis] Hydrolysis uses water molecules to cleave covalent bonds within polymers, breaking them down into constituent monomers.
35	B	lipids	[Four Major Biomolecules] The four primary categories of biomolecules are carbohydrates, proteins, lipids (fats), and nucleic acids.
36	D	carbohydrates	[Carbohydrates Overview] Carbohydrates (sugars and starches) are made of C, H, and O and function primarily as accessible energy sources for the body.
37	B	monosaccharides	[Monosaccharides] Monosaccharides, or simple sugars, are single-ring sugar molecules containing 3 to 7 carbon atoms.
38	A	hexoses	[Hexose Sugars] Hexoses are monosaccharides containing six carbon atoms (C ₆ H ₁₂ O ₆), including glucose, fructose, and galactose.
39	A	glucose	[Glucose] Glucose is the essential blood sugar hexose utilized directly by cells for cellular respiration.
40	D	fructose	[Fructose] Fructose is a fruit sugar hexose found predominantly in fruits and vegetables.
41	A	galactose	[Galactose] Galactose is a six-carbon hexose prominent in dairy products.
42	A	ribose	[Ribose] Ribose is a 5-carbon pentose sugar found in ribonucleic acid (RNA) and ATP.
43	D	oligosaccharides	[Oligosaccharides] Oligosaccharides are carbohydrate polymers composed of a small number of linked monosaccharide units.
44	A	disaccharide	[Disaccharides] Disaccharides are formed by covalently joining two monosaccharides through dehydration synthesis.
45	C	sucrose	[Sucrose] Sucrose is a disaccharide formed from one glucose and one fructose molecule.
46	C	polysaccharides	[Polysaccharides] Polysaccharides are large polymers made of many monosaccharide subunits.

47	A	starch	[Starch] Starch is the primary energy storage polysaccharide synthesized by plant cells.
48	A	glycogen	[Glycogen] Glycogen is the animal storage form of glucose, found in abundance in liver and muscle tissue.
49	D	cellulose	[Cellulose] Cellulose provides structural rigidity in plants and cannot be digested by human enzymes, acting as dietary fiber.
50	D	lipids	[Lipids Overview] Lipids are nonpolar organic molecules that do not dissolve in polar solvents like water.
51	D	triglycerides	[Lipid Classes] The three major human lipid classes are triglycerides (fats/oils), phospholipids, and steroids.
52	C	fatty acids	[Triglyceride Structure] Triglycerides consist of a single glycerol backbone esterified with three fatty acid tails.
53	B	saturated	[Saturated Fatty Acids] Saturated fatty acids possess only single carbon-carbon bonds, maximizing the number of bonded hydrogen atoms.
54	D	unsaturated	[Unsaturated Fatty Acids] Unsaturated fatty acids feature at least one double bond in their carbon chain, introducing kinks in the tail.
55	C	twice	[Energy Density of Fats] Triglycerides store roughly 9 kcal/gram, more than twice the energy per gram of carbohydrates or proteins (4 kcal/gram).
56	D	amphipathic	[Phospholipid Amphipathic Nature] Amphipathic molecules feature both water-attracting (hydrophilic) and water-repelling (hydrophobic) regions.
57	A	phospholipid	[Phospholipids in Membranes] Phospholipids self-assemble into a bilayer that forms the structural basis of cellular plasma membranes.
58	B	steroids	[Steroids] Steroids are lipid molecules recognized by their characteristic four fused carbon rings.
59	B	cholesterol	[Cholesterol] Cholesterol maintains membrane fluidity and serves as the biochemical backbone for steroid hormones.
60	A	proteins	[Proteins Overview] Proteins are versatile biopolymers made from chains of amino acid residues.
61	C	amino acid	[Amino Acids] Amino acids are the organic monomers that link together to synthesize polypeptide chains.
62	A	carboxyl	[Amino Acid Structure] Amino acids have a central alpha-carbon bonded to -H, an amino group (-NH ₂), a carboxyl group (-COOH), and an R-group.

63	C	side chain	[Amino Acid Side Chain (R Group)] The side chain (R-group) varies among the 20 amino acids, conferring specific charge, polarity, and reactivity.
64	D	non-essential	[Non-Essential Amino Acids] There are 11 non-essential amino acids that the human body can synthesize on its own.
65	B	essential	[Essential Amino Acids] The 9 essential amino acids cannot be synthesized by human cells and must be ingested.
66	A	primary	[Primary Protein Structure] Primary structure is the unique linear sequence of amino acids, as seen in hormones like insulin.
67	B	secondary	[Secondary Protein Structure] Secondary structure refers to localized folding into alpha-helices or beta-pleated sheets.
68	C	tertiary	[Tertiary Protein Structure] Tertiary structure is the full 3D spatial conformation of a single folded polypeptide.
69	D	quaternary	[Quaternary Protein Structure] Quaternary structure exists when a protein is composed of multiple interacting polypeptide subunits (e.g., hemoglobin, DNA polymerase).
70	A	enzymes	[Enzymes Overview] Enzymes are protein catalysts that lower activation energy, speeding up metabolic reactions without being consumed.
71	B	substrate	[Enzyme Substrates] The substrate is the specific molecule that binds to an enzyme's active site to undergo chemical alteration.
72	D	active	[Enzyme Active Site] The active site is the catalytic pocket where substrate molecules fit precisely like a lock and key.
73	B	cofactors	[Cofactors] Cofactors are nonprotein helper components necessary for enzyme function.
74	A	coenzymes	[Coenzymes] Coenzymes are organic cofactors (often derived from vitamins) that facilitate catalytic transfer during reactions.
75	A	DNA	[Nucleic Acids Overview] Deoxyribonucleic acid (DNA) and Ribonucleic acid (RNA) are the informational polymers of life.
76	B	nucleotide	[Nucleotides] Nucleotides are the monomers that polymerize to form DNA and RNA chains.
77	B	gene	[Genes] A gene is a hereditary unit of DNA containing the genetic code for assembling a polypeptide.
78	B	deoxyribose	[Deoxyribose Sugar] DNA contains deoxyribose sugar, which lacks an oxygen atom on carbon 2 compared to ribose.
79	B	thymine	[DNA Bases (Thymine)] Thymine (T) is found exclusively in DNA; RNA

			replaces thymine with uracil (U).
80	A	RNA	[RNA Overview] Ribonucleic acid (RNA) is typically single-stranded and features ribose sugar and uracil.
81	A	uracil	[RNA Bases (Uracil)] In RNA, uracil (U) takes the place of thymine and pairs with adenine during transcription.
82	D	protein	[Role of RNA in Protein Synthesis] RNA molecules (mRNA, tRNA, rRNA) orchestrate protein synthesis by translating genetic instructions into polypeptide chains.